

Danni Liu

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EDUCATION

- PhD** **The University of Chicago** | Chicago, IL | Department of Computer Science June 2028 (Expected)
Daniels Fellowship
- BS** **Washington University in St. Louis** | St. Louis, MO | Department of Computer Science May 2023
Cum Laude

RESEARCH SUMMARY

- I design and study interactive, AI-powered systems that help people explore, interpret, and make decisions with complex data, integrating human-centered research with interactive system design, development, and empirical evaluation.

RESEARCH EXPERIENCE

Chi Data Group, The University of Chicago Chicago, IL
Graduate Researcher October 2025 – Present

- Designed and conducted survey, interview, and controlled user studies on how data workers search for, evaluate, and verify datasets and LLM-generated answers under uncertainty
- Analyzed qualitative and behavioral data to identify resource-rational evaluation strategies and translate findings into design implications for AI-assisted data discovery systems
- Designed and implemented multiple explanation mechanisms in the system and conducted controlled user studies to evaluate their impact on user trust calibration and interaction behavior

Data Cognition Lab, The University of Chicago Chicago, IL
Graduate Researcher September 2023 – October 2025

- Led a mixed-methods study with 41 data workers to characterize breakdowns in data discovery workflows, revealing systematic reliance on heuristic and socially derived signals, informing the design of data marketplace interfaces
- Developed a research prototype of a task-aware dataset recommendation system enabling users to express analytical intent, focus ranges, and salient terms for exploratory data analysis

Visual Interface & Behavior Exploration Lab, Washington University in St. Louis St. Louis, MO
Undergraduate Researcher May 2022 – August 2022

- Developed a visual analytics system with University of Colorado collaborators to explore interactive approaches for comparing chemical compositions and synthesis pathways in material discovery

RealAI Beijing, China
User Research Intern May 2021 – August 2021

- Conducted semi-structured interviews with 21 computer vision engineers to model practitioner understanding of adversarial vulnerabilities, informing the design of an educational system for robustness awareness

PUBLICATIONS

Toward Resource-Rational Dataset Search Interfaces.

Liu, D., Xia, S., Castro Fernandez, R. and Kale.

DASHSys 2026: Workshop on Systems for Data-Centric Agents with Human-in-the-Loop, co-located with VLDB 2026. Forthcoming.

What Can Interactive Visualization Do for Participatory Budgeting in Chicago?

Kale, A., Liu, D., Ayala, M. G., Schwab, H., & McNutt, A.

IEEE Transactions on Visualization and Computer Graphics (TVCG), 2024.

Music-to-Facial Expressions: Emotion-Based Music Visualization for the Hearing Impaired.

Wang, Y.*, Pan, F.*, Liu, D.*, & Hu, J. (*Equal contribution.)

AAAI Conference on Artificial Intelligence (AAAI), 2023.